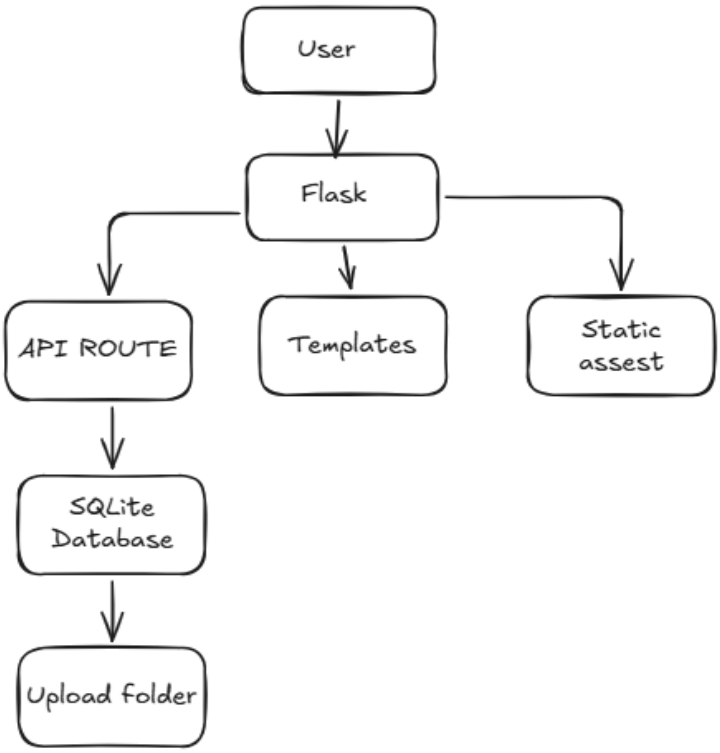
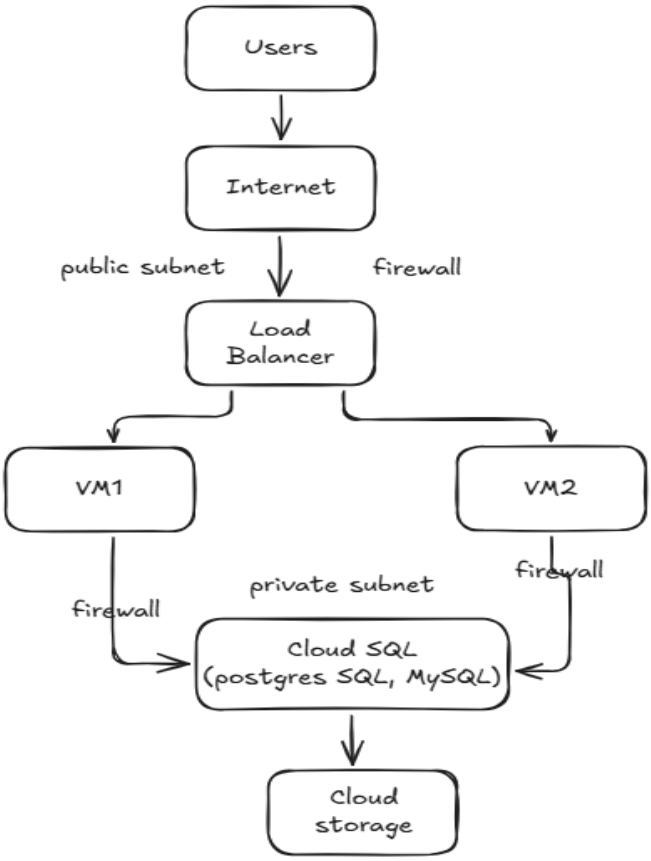
CloudNotes Architecture Proposal

Screenshot 1 – Current Local Architecture



CloudNotes currently runs on one local machine using Flask, SQLite, templates, static assets and uploads. This design is suitable for development but creates a single point of failure and cannot scale.

Screenshot 2 – Proposed Cloud Architecture



Only the Load Balancer is public. Compute Engine VMs and Cloud SQL are placed in private subnets protected by firewall rules and IAM. Cloud Storage stores uploaded files.

Screenshot 3 – Component-to-Service Mapping

Component	Category	GCP Service	Justification
Flask App	Compute	Compute Engine	Runs application logic
API Routes	Compute	Compute Engine	Served by Flask
Templates	Compute	Compute Engine	Rendered by Flask
Static Assets	Storage	Cloud Storage	Durable object storage
SQLite	Database	Cloud SQL (PostgreSQL)	Managed HA database
Upload Folder	Storage	Cloud Storage	Persistent uploads
User Requests	Networking	Cloud Load Balancer	Distributes traffic
Security	Security	IAM + Firewall	Identity and network protection

Screenshot 4 – Distributed Systems Risk Analysis

Risk	Mitigation
DB zone failure	Cloud SQL HA standby

10x traffic	Autoscaling + Load Balancer
VM failure	Multiple VMs
Storage failure	Cloud Storage
Network attack	Private subnet + IAM + Firewall

Screenshot 5 – Architecture Decision Record

• Decision: Use Cloud SQL instead of SQLite. Trade-off: Higher cost for backups, durability and HA.

• Decision: Use multiple Compute Engine VMs behind a Load Balancer. Trade-off: Higher cost for scalability and fault tolerance.

• Decision: Store uploads in Cloud Storage. Trade-off: Additional storage cost for durable object storage.

• Decision: Keep backend resources in private subnets protected by IAM and firewall rules.